

Bulletproofs:

Short Proofs for Confidential Transactions and More

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- 1 Motivation
- 2 Range Proofs
- 3 Bulletproofs

Bitcoin is neither confidential nor anonymous

Summary

Size	1110 (bytes)
Fee Rate	0.0016173243243243243 BTC per kB
Received Time	June 16, 2026 12:38:00 AM
Mined Time	June 16, 2026 12:38:00 AM
Included In Block	00000000000000000f0115cca58564683b337404832c88539ce2995e799e5c

Details

c2561b292ed4878bb28478a8cafd1f99a01faeb9c5a906715fa595cac0e8d1d8

mined June 16, 2026 12:38:00 AM

Inputs

16k436SRzdeCPKGwJDNNBEkXj696MbChwx

0.53333328 BTC

1Bsh4KD9ZJT4dJcoo7SSuS1jvtmtVmREb7

1.47877788 BTC

Outputs

1JgVBpw5TDMTr0ZXg9XpPDQRRHtNbSCsPA

0.01031593 BTC (U)

1AFLhD4EtG2uZmFxmfdXCyGUNqCqDS887u

2 BTC (S)

FEE: 0.00179523 BTC

1 CONFIRMATIONS

2.01031593 BTC

Figure: Bitcoin transaction scheme

Validity of a Bitcoin Transaction

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- $\sum (\text{input values}) = \sum (\text{output values}) + \text{fee}$

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- Imagine Gebze Technical University pays academicians in Bitcoin.

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- Imagine Gebze Technical University pays academicians in Bitcoin.
⇒ All salaries are public

Solution: Confidential Transactions [Maxwell 2015]

Idea: replace “value” by commitment to “value”

Pedersen commitment:

$$\text{commit}(v) = g^v \cdot h^r$$

(random r)

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$$g^{533} h^{r_1}$$

1JgVBpw5TDMTRoZXg9XpPDQRRHtNbSCsPA

$$g^{10} h^{r_3}$$

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Amounts no longer written in blockchain !!!

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To verify blockchain, check that for all transactions:

$$\sum (\text{input values}) = \sum (\text{output values}) + \text{fee} \quad (*)$$

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Solution: payer attaches **zero knowledge proofs** to transaction

- 1 (*) holds for committed values, and
- 2 all committed outputs are **positive** in $[0, 2^{52})$

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public on blockchain: $\text{commit}(v) = g^v \cdot h^r$

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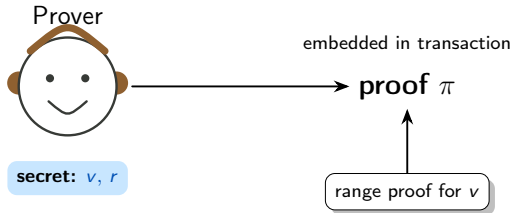
Prover



secret: v, r

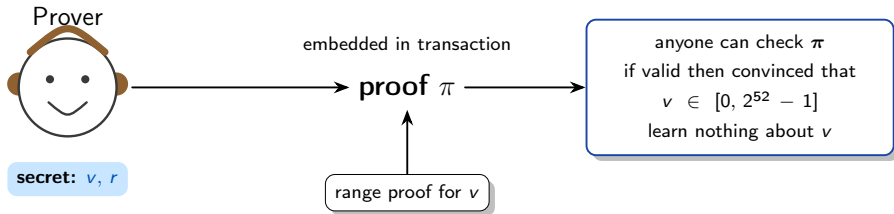
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aka Range Proof $v \in [0, 2^{52} - 1]$



Alice



Bob

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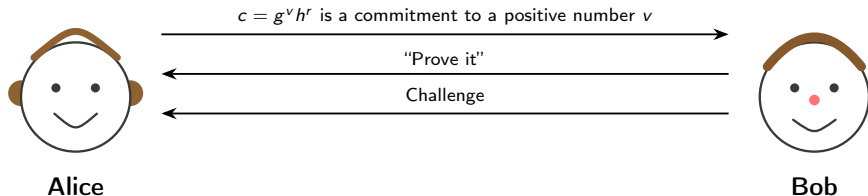
$c = g^v h^r$ is a commitment to a positive number v



Bob

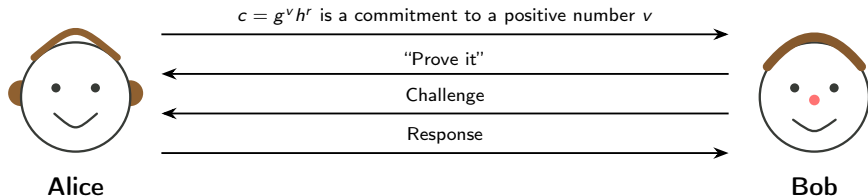
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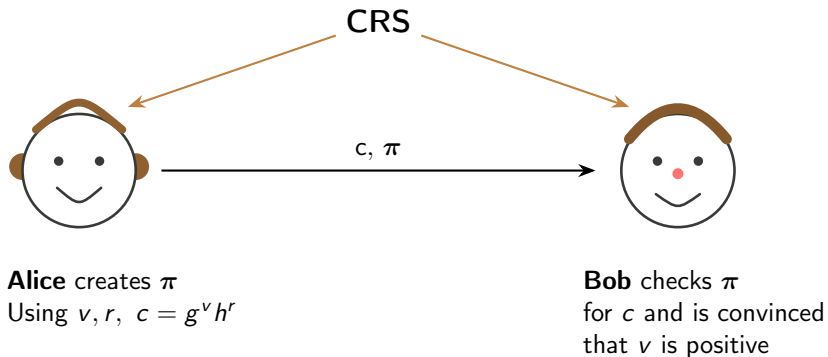


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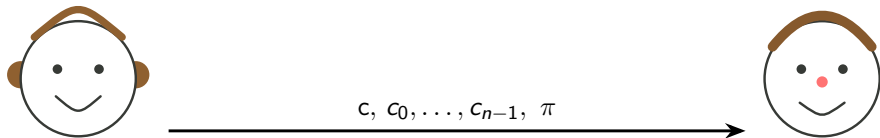
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Non-Interactive Zero Knowledge Proof of Knowledge (NIZK)



Concrete Range Proof using bit commitments



$$v = (b_0, \dots, b_{n-1}), \quad b_i \in \{0, 1\}$$

$$r_i \leftarrow \mathbb{Z}_q \quad \forall i \in [0, n-1]$$

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Concrete Range Proof using bit commitments

$$c_i = \text{commit}(b_i, r_i) \wedge v = \sum_{i=0}^{n-1} b_i \cdot 2^i \wedge b_i \in \{0, 1\}$$



$c, c_0, \dots, c_{n-1}, \pi$



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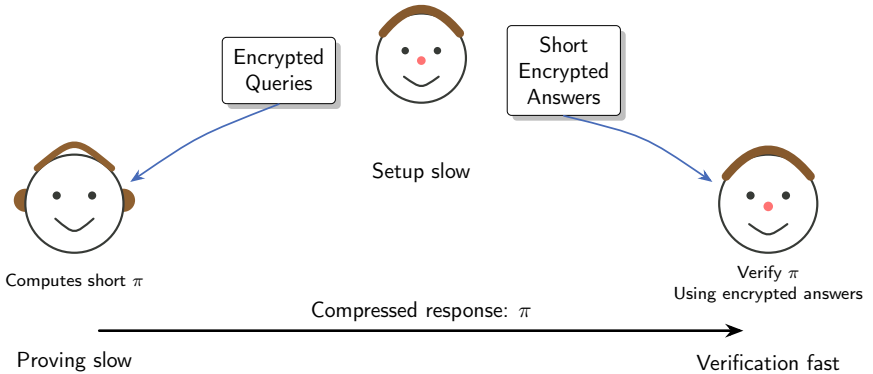
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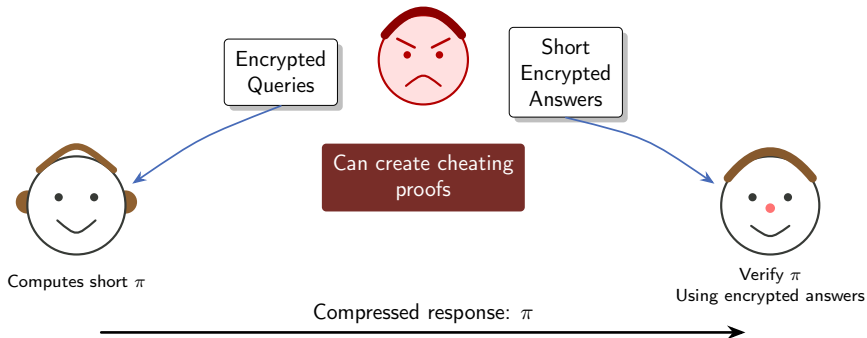
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- No **trusted setup**

Preprocessing SNARKs with Trusted Setup (GGPR 13)



Preprocessing SNARK: Malicious Setup



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- **Problem: trusted setup**
 - Reguire a public large string($\approx GB$) generated in secret
 - If subverted, prover can create false proofs
 - $\Rightarrow$ undetectable inflation
 - Even fear of undetectable inflation is problem.

General proofs on committed values

- Builds on the system of [Bootle et al. 2016]

n-bit range proof size: $\approx \log(n)$!!!

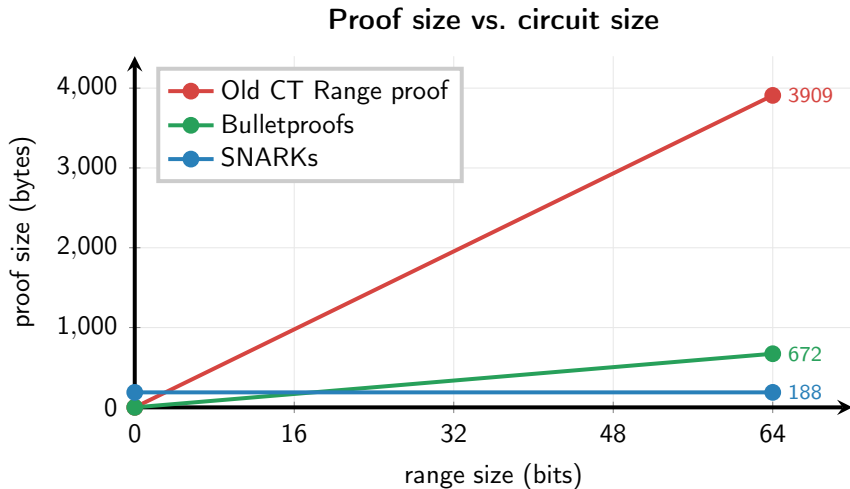
- Compare to $\approx n$ previously (Exponential reduction)
- For t proofs: $\approx \log(n) + \log(t)$

No trusted setup

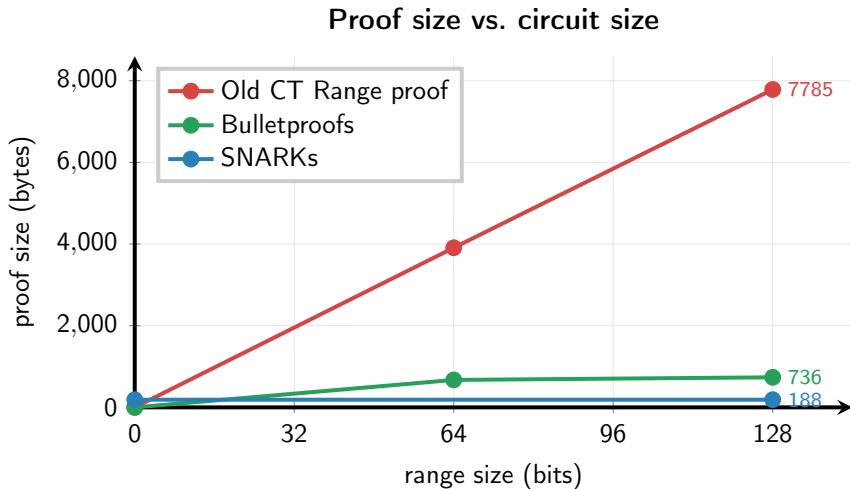
Bulletproofs for Confidential Transactions/Mimblewimble

- 670 bytes instead of 4KB per range proof (64 bit range)
- Aggregation: Two range proofs 736 bytes vs. 8KB
- 16 range proofs 928 bytes vs 61 KB
- Doubling precision adds 64 bytes
- UTXO set/Mimblewimble size: 17 GB v 160 GB
- Built in: simple Coinjoin protocol for combining confidential transactions
- No unconditional soundness quantum soundness

Bulletproof vs 1 Range Proof

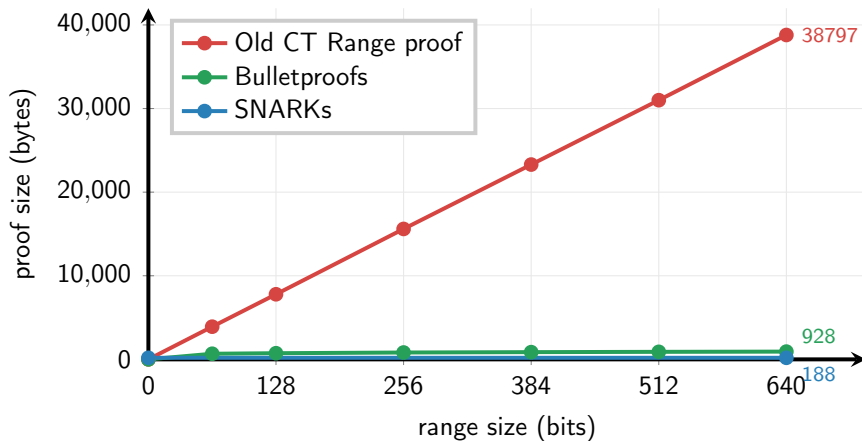


Bulletproof vs 2 Range Proof



Bulletproof vs 10 Range Proof

Proof size vs. circuit size





Benedikt Bünz, Jonathan Bootle, Dan Boneh, Andrew Poelstra, Pieter Wuille, Greg Maxwell, *Bulletproofs: Short Proofs for Confidential Transactions and More*, 2018.



Greg Maxwell, *Confidential Transactions*, 2015.

[https:](https://elementsproject.org/features/confidential-transactions)

[//elementsproject.org/features/confidential-transactions](https://elementsproject.org/features/confidential-transactions)



Jonathan Bootle, Andrea Cerulli, Pyrros Chaidos, Jens Groth, Christophe Petit, *Efficient Zero-Knowledge Arguments for Arithmetic Circuits in the Discrete Log Setting*, 2021.

Thank you!
Questions?